Continuity of Care Document for Hospital Management Systems: an Implementation Perspective

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ABSTRACT
Implementation of Electronic Health Record (EHR) has become essential towards delivery of standardized healthcare services, particularly in developing nations. Legacy hospital management systems developed earlier lack compliance to EHR guidelines and required standards. In this paper, we focus on a specific aspect of EHR, the Continuity of Care Document (CCD), which is a standard format used for ubiquitous migration of patient information and medical records across hospital management systems. We discuss the challenges involved in implementation of CCD for legacy systems and present case studies on such a system with a view to understand the gap and effort required in such an endeavour. We also introduce an SDK and web API for assisting organizations to easily extending the existing Hospital Information Systems to generate CCD. This would be useful for developers of healthcare solutions in understanding and practically implementing the same.

CCS Concepts
• Applied computing~E-government

Keywords
Hospital Management Information Systems; Continuity of Care Document; Electronic Health Record; Health Standards

1. INTRODUCTION
One of the key challenges facing implementation of the Electronic Health Record (EHR) in countries such as India has been the lack of implementation of health standards [27]. With increasing number of medical facilities and a flourishing medical tourism industry, compliance to standards is crucial as it ensures uniformity and ubiquity in storage and transmission of patient health data among different systems.

The Continuity of Care Document (CCD) is an eXtensible Markup Language (XML) based specification for standardizing the storage and transfer of summary of patient records [23]. The primary point of care for a majority of patients is clinics, general hospitals and laboratories [31]. Therefore, by the time the patient needs advanced care, he has already gone through a lot of medication and tests. Usually these tests are communicated to the advanced medical facility via printed reports [30]. In a few cases, the primary point of care provides a clinical summary for the patients.

This may result in several problems. Since there is no standardized format for generation of documents pertaining to administered treatment, investigation reports, medications prescribed / administered or discharge summaries, interoperability in the presence of lack of standardization presents a challenge. Further, there is difficulty in tracking the patient history as it may not be easily available. Moreover, crucial time may be lost in case of emergencies in the process of collation of medical records, which has to be avoided.

In this paper, we analyze the CCD from the perspective of the challenges faced and learning outcomes in implementing the same for existing Hospital Management Information Systems (HMIS). In order to motivate the context of the paper, we review available literature in the context of health standards' implementation.

Lee et al.[15] present an HMIS on cloud for CCD generation and integration. Cook et al. [5] characterize gaps in continuity of clinical care as a result of organizational boundaries, loss of information during patient transfer or other care based processes. They opine that these can be resolved via suitable technical measures that are developed to minimize human intervention. Haggerty et al. [11] categorize continuity into three types: informational, management and relational. A statistical evaluation of the effect of delay in continuity has been presented in the work by Kriplani et al. [12] which concludes that the same can critically affect patient care and standardization is required for ubiquitous provisioning of patient records. As a consequence, several standards have been developed with the objective of uniformly representing patient care information for interoperability between systems.

Ferranti et al. [9] compare and contrast the Clinical Document Architecture (CDA) and the Continuity of Care Record (CCR) while emphasizing upon the mutual compatibility among standards. Laleci et al. [13] present scalable semantic framework for interlinking clinical research such as MedDRA, WHODD and CDISC and clinical care domains (SNomedCT, LOINC, ICD-10 [22]). Barca et al. [3] develop the *yourEHRM* to address the issue of interoperability using a dual model approach; they use a combination of a domain invariant reference model and specific clinical models to achieve this.

DÁmore et al. [6] stress the importance of Continuity of Care Document (CCD) as an effective interoperability standard. An empirical evaluation of the CCD with Rx-Norm for Clinical Drug Representation has been represented by Linas et al. [16]. Our working environment presents a scenario where there are multiple Hospital Management Information Systems (HMIS) which are being independently used at medical facilities for storage and management of patient information and activities. However, since the standards are relatively new, these were not in place when these systems were developed. In the present day scenario where EHR implementation has become a necessity, the implementation of health standards in such systems presents a challenge. A few of the challenges include lack of information (missing data fields) as these were not captured in the system, disparate databases and schemas, multiple operating environments and associated constraints. Based

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on these challenges, we present a case study for implementation of the CCD for such a scenario and the challenges faced in the same.

The rest of the paper is organized as follows. Section 2 introduces the CCD and presents its basic structure. The challenges faced in implementation of the CCD along with the learning outcomes are discussed in Section 3. In order to accelerate the development effort required to make an HMIS CCD compliant, we develop an SDK that provides an easy abstraction for data acquisition and CCD generation, which is discussed in Section 4. Finally, conclusions and future work are presented in Section 5.

2. CCD DOCUMENT STRUCTURE

The CCD document consists of sections characterizing various aspects of patient care information. These are categorized into mandatory and optional sections, as shown in Fig. 1. A CCD document is a valid document if it has all the mandatory sections. These include allergies, medications, problem list, procedures and results. Optional sections include advance directives, encounters, family history, social history, functional status, immunizations, medical equipment, payers, plan of care and vital signs. Optional sections may or may not be included depending on the availability of data in the existing HMIS systems.

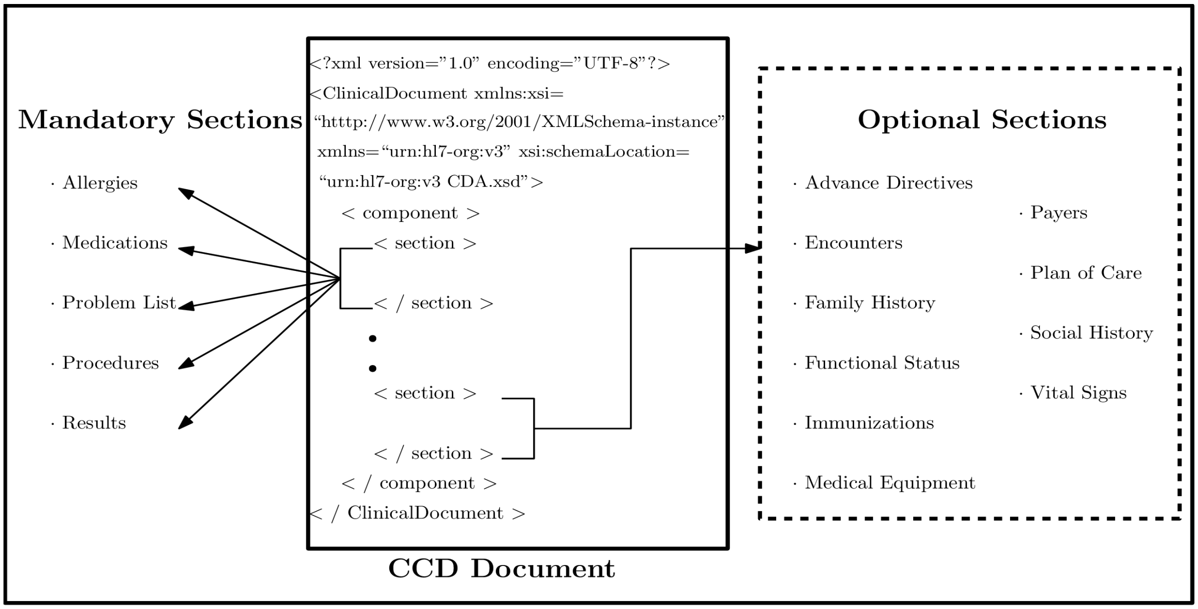

Figure 1: Overview of the structure of the CCD document

2.1 Medication

The medication section of the CCD document contains details about the current as well as history of medications prescribed and used by the patient. These are represented by the *medication* and *supply* activities. It should mandatorily contain at least the patient's current medications. The *medication* activity contains details of the medicines prescribed while the *supply* section is concerned with what has actually been dispensed.

As the patient may not necessarily take the same dosage or presentation of a medicine prescribed by the clinician, *medication* and *supply* activities are expressed in the "EVN" (Event) or "INT" (Intent) mood. The "EVN" mood reflects actual use while the "INT" mood indicates what the clinician intends the patient to take. *Medication* section, characterized by *SubstanceAdministration*, primarily consists of the following:

1. An identifier and status code.
2. Time elements for start and stop of medication.
3. Route of administration of medication.
4. Dose quantity (and/or maximum dose quantity).

The *supply* section, which contains details of dispensing activities, consists of the following:

1. An identifier and status code.
2. Effective time to indicate time of dispensing.
3. Repeat number indicating number of fills.
4. Quantity and author (prescriber) information.

A significant challenge presented in implementation of the medication section is the format used to represent the actual medication. Examples of clinical vocabularies include RxNorm [21, 17], SNOMED CT [14, 8], LOINC [18, 7] among others. Typically, CCD uses RxNorm code system for medications, and the CDC Vaccine Code system [1] for immunizations.

2.2 Results

The results section of the CCD constitutes of information regarding the following categories of patient data:

- Laboratory investigations (eg. biochemistry results)
- Imaging results (eg. imaging investigations such as ECG. X-ray, MRI etc.)
- Procedure results (specific granular information generated by clinicians during specific procedures not accounted as the above categories)

The results section is represented by two components, result *organizer* and *observation*. *Observations* are contained within an *organizer*. An *organizer* consists of an identifier, status code, code (selected from LOINC or SNOMED CT, specimen (as in procedure) and component (to indicate technique by which result is obtained). The result *observation* represents the actual test result, with an identifier, status code, effective time, method code, observation values, reference range and source of information. Data for the results section is usually populated from the investigation module of the HMIS.

2.3 Problem List

The problem list section consists of information regarding all clinical problems of the patient at the time of generation of the CCD. Problems are represented as activity (*act*) or *observation*. *Act* includes observations which can be a problem or alert observation. For an observation, the mood code, status code, effective time, code and source are captured. Multiple occurrences of problems are represented as episode observations, which have their distinct identifiers. Patient awareness of a problem is captured by the *participant* block.

2.4 Allergies, Adverse Reactions and Alerts

This section lists the patient's current and historic allergies, adverse reactions to specific substances and alerts. These are represented similar to the previous section, as *act* or *observation*. Alert observations include mood code, status code, effective time, value and source. The entity causing the respective allergy or adverse reaction is represented as the agent, using a *participant*. *Observations* can also be reaction or severity observations, or reaction interventions.

2.5 Procedures

The procedures section comprises of details about all *"interventional, surgical, diagnostic, or therapeutic procedures"* administered to the patient. It is represented as an activity (*act*), *observation* or *procedure*. The identifying attributes are an ID, status code, effective time, procedure code (from among the LOINC, SNOMED-CT, CPT-4 [2], ICD-9 [10] or ICD-10 coding systems). Additional attributes include the target site code (to indicate the anatomical position of the procedure), location participants (includes participant role and/or playing device), performer (to identify the assigned entity or organization to perform the procedure), authorization, specimen and/or source.

2.6 Optional CCD Sections

In this section, we briefly discuss the optional sections in the CCD for the sake of completeness of the overview of the structure of the CCD. Depending on data availability in the HMIS systems, these sections may be considered to be implemented in the CCD.

2.6.1 Encounters

The encounters section in the CCD document is used for keeping track of the number of visits made by the patient with the consulting physician or a similar entity. These include inpatient/outpatient visits, emergency visits, ambulatory, home, field, daytime or virtual encounters. Encounters are represented with their category, time stamp and supporting details such as the contact information of the encounter participant.

2.6.2 Family History

The family history section contains the details of the patient's genetic relatives in terms of potential risks to the patient's health profile. The history is organized to reflect observations and organizers. In case where the aspects pertaining to family history are related to death, they are specifically recorded as a "cause of death" observation.

2.6.3 Social History

This section of the CCD document contains information about the patient's lifestyle, environmental and administrative aspects. Typical examples include marital status, religious affiliation, race, ethnic group etc. Entries are represented by the observation class and contain an identifier, description, status code and effective time attributes. The social history section information is often unavailable in existing hospital management systems unless the medical practitioner would have collected them as part of diagnosis procedures. The availability of this information is often determined as the standard compliant system continues to be used over time.

2.6.4 Vital Signs

This section comprises of information about the patient's vitals such as height, weight, blood pressure, temperature, pulse rate etc. These are mostly available when these are captured in the system prior to a consultation appointment with the practitioner or for the case of patients undergoing treatment in a medical facility.

2.6.5 Immunizations

This section contains information about the patient's current immunization status and the previous immunization history. It is recommended to be included in the CCD for cases pertaining to pediatric care. The structure of the immunization section is similar to that of the medication section (using the *SubstanceAdministration* class), and follows the Center for Disease Control and Prevention (CDC's) CVX (vaccine administered) code set [1].

2.6.6 Medical Equipment

The medical equipment section contains details about internally implanted or external medical equipment that the patient uses or that affects the patient's health status, along with similar history. The information is described using the *supply* class, similar to that in the medication section. A few challenges in interfacing medical equipments with HMIS with a view of interoperability has been presented in [26].

2.6.7 Payers

Financial aspects of the patient's care regime are included in the payers section which includes insurers, guarantors, and other empaneling entities which may support patient care financially. This may include activities of *coverage*, *policy* or *authorization*. *Coverage* activities may include multiple *policy* activities, each of which may contain *authorization* activities.

3. CHALLENGES IN IMPLEMENTATION

Figure 2 summarizes the interaction between two HMIS systems which have the EHR of a patient via the CCD. It serves as an interface for communication of information related to patient care between the systems which may have independent EHRs pertaining to the same patient. The systems can independently parse the CCD to obtain standardized representation of patient care information which can then be updated in the local patient EHR.

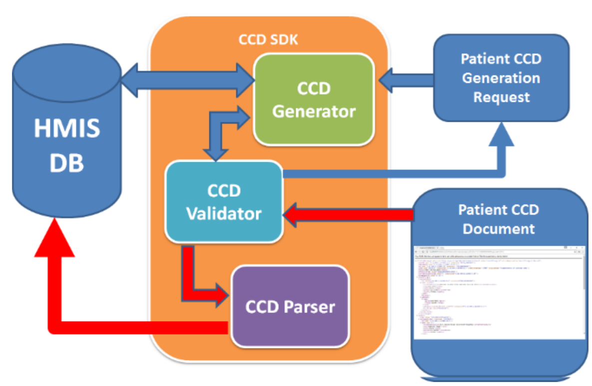

Figure 3: CCD SDK architecture and working

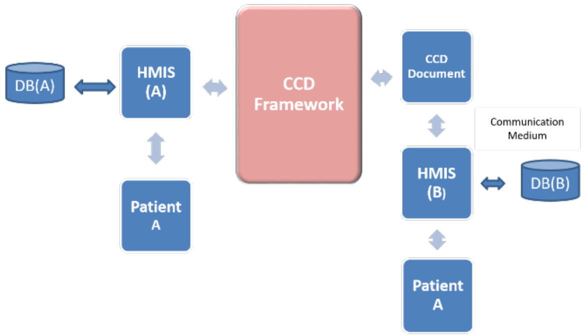

Figure 2: HMIS systems interfacing patient health data through the CCD

One of the issues faced in implementing the CCD for disparate HMIS systems is the identifier used for patient identification. Independent systems use different identifier generation schemes, and they also may have redundant patient entries owing to the nature of the data entry processes employed for such systems in practice. A standardized and robust identifier generation mechanism should be employed and enforced in HMIS systems prior to implementation of health standards. One such scheme is presented in [25] which uses encoding of patient attributes to generate identifiers with the added advantage of facilitating duplicity detection.

There are various challenges presented by the availability of data required to generate the CCD document, in terms of data formats, uniformity, redundancy and inconsistencies. Often, data required for the mandatory sections of the CCD document may not be available in the existing HMIS. There may be multiple formats in which data of the same kind is represented, for example timestamps, medicine quantities etc. may not be conformant to the standard unit systems. Many data attributes may not even be captured by the systems. Addressing data-related challenges is a key component from the implementation section [24, 28].

A significant challenge is presented in the mapping of the data attributes required to generate the CCD sections as independent HMIS systems store patient care data in their respective databases using uncorrelated schemas. As such, extracting the required data requires manual development efforts in terms of creation of data mappers that can fetch the data from the databases of the HMIS to generate the CCD. This process can often be time consuming, error-prone and tedious. Umer et al. [29] and Menezes et al. [19] present a mechanism to address this issue for the HL7 specification implementation where they attempt to find the closest semantic match to the required set of attributes by searching a mapping knowledge repository. This mechanism is specifically beneficial when multiple systems need to be made CCD compliant. In the following section, we discuss the various modalities available for CCD generation and demonstrate the benefit of the SDK developed by us for this purpose.

4. SDK FOR CCD GENERATION

A major bottleneck faced in practical implementation of CCD is the absence of any model implementation. Conventional ways in which the CCD can be generated are summarized in Table 1.

These tools require the developer to be well-versed with the CCD architecture and underlying data types (for eg. TS, ST) which are common among the different CCD sections. MDHT and other tools attempt to overcome this by providing an abstraction for various sections. However, the user is still required to construct the basic data objects for the data types and set the data from the respective fields in the database schema while generating the CCD. In order to overcome these limitations, we developed a Software Development Kit (SDK) for accelerating the CCD generation process. The SDK has the following components (shown in Fig. 3):

- CCD generator: Allows for mapping of CCD attributes with database schemas for CCD generation. The primary functionality of this component is to map the attributes from the HMIS database to the attributes specified in the CCD standard. The minimum attribute set for the mandatory sections of the CCD need to be provided to ensure that a valid CCD is generated, while additional optional sections along with their attributes can also be mapped and populated, depending upon their availability in the HMIS. This component addresses the challenge of disparate data types present in different systems as well as taking care of the mapping of these data attributes to the CCD. It provides for configurable data mappers that can be customized during implementation according to the nature and format of availability of data in the prevailing HMIS. The developer need not write code for explicitly implementing these maps, which saves on significant development time and effort.

- CCD parser: Parses the data from CCD to formats for processing at the EHR. This component is responsible for doing the inverse operation performed by the CCD generator. When a patient's CCD is received, this component is responsible for parsing the individual CCD sections, extracting the patient's clinical data and information encoded in the CCD and populating the database of the corresponding EHR/HMIS system with the patient data. This is further consumed by the system for the processes or functions it performs on the patient data.

- CCD validator: This component is responsible for validating the generated CCD. It serves as a sanity check on the generated CCD document. In order for the CCD generated from an HMIS system to be compliant to the standard specifications, it has to pass certain validation criteria which include:

- CCD Base Standard Only

- Base Standard Plus Templated Validation, which may include HL7 Balloted Implementation Guides, Healthcare Associated Infections (HAI) (HL7 Balloted IGs), Quality Reporting Document Architecture (QRDA), Consolidated CDA (C-CDA) (HL7 Balloted IGs) etc.

The CCD validator component checks that the generated or received CCD document is valid in accordance with the specified criteria, failing which the CCD document is rejected. Upon successful validation, the CCD document may be transmitted to another EHR system (when generated) or sent to the CCD parser (when received).

The SDK serves two use-cases. The first is the case for generation of a patient's CCD from an HMIS. In this case, a request to generate the CCD is sent from the user's interface to the CCD generator. The CCD generator communicates with the HMIS database via queries to get the data required to generate the CCD. Once this is accomplished, the CCD is validated using the CCD validator module and communicated back to the request generator. The second use case is when a patient's care information is available in the form of a CCD and needs to be fed into an EHR in an HMIS. In this case, the CCD is sent to the CCD validator which checks whether the received document is a valid CCD or not. It is then sent to the CCD parser which extracts the relevant information and parses them to be saved into the HMIS database.

As discussed previously, the CCD generation process also entails the creation of a patient identifier, which needs to be unique for each patient. We use the scheme proposed in [25] which involves the generation of patient identifiers using Soundex encoding of a combination of patient attributes. The patient attributes used include the patient name, patient’s father’s/mother’s name, date of birth, gender, state and district of residence. The Soundex encoding of the name attributes is used and the geographic attributes are encoded using the metadata standards. In order to ensure the uniqueness of the generated identifier, a 3-character alphanumeric sequence is appended at the end of the identifier, which allows for generation of upto 46,656 unique identifiers for patients with similar attribute information. This scheme has been benchmarked for robustness on a patient registration dataset in the work presented in [25].

The attributes required for the generation of the CCD have been provided as classes with required methods for data parsing in the SDK, with an objective to reduce the development effort in CCD generation. To give an example, we compare the use of MDHT against our SDK for populating the address field in Fig. 4. One can see that there is a significant reduction in the number of lines of code to be written to accomplish the same task, which will save on development effort. The SDK can be configured to interface with the EHR database and fetch the required patient details to generate the CCD. Moreover, since the SDK is developed with a high level of abstraction, a developer need not to be familiar with underlying data types as represented in CCD reference schema. As a result of the SDK, we observed an average reduction in lines of code by around 50% when compared with MDHT. This can potentially save crucial man-hours when compared to the conventional methods that also have a steep learning curve.

The class diagram for a module of the SDK used for generation of patient information for CCD is shown in Fig 5.

MDHT:

```
providerOrganizationAddr.setCity("city");
providerOrganizationAddr.setState("state");
providerOrganizationAddr.setCountry("country");
providerOrganizationAddr.setPostalCode("XXXXXX");
providerOrganizationAddr.setDeliveryAddressLine("deliveryAddress");
providerOrganizationAddr.setHouseNumber("houseno");
providerOrganizationAddr.setStreetName("streetname");
objProviderOrganization.getAddr().add(providerOrganizationAddr);
```

Our CCD SDK:

```
providerOrganization.addAddress(db.getProviderOrganizationAddress());
```

Figure 4: CCD address field population using MDHT compared to our SDK

5. CONCLUSION AND FUTURE WORK

This paper presented an implementation viewpoint for the enforcement of CCD for storing patient health data and interoperability of EHR systems. Various sections of the CCD document have been discussed and challenges presented in their implementation have been highlighted. In order to speed up the implementation and reduce development effort, we created an SDK that provides entity interfaces and methods for the developer to easily configure the same to interact with the database schema and generate the CCD document. Since present day HMIS are in their infancy with regard to conformance to health standards, the impact analysis of these implementations remains to be seen as time progresses and the CCD becomes prevalent as a modality for patient information exchange. Once standards such as CCD have been implemented, studies on feasibility analysis and effectiveness in continuity of patient care need be conducted and their true benefit ascertained.

Table 1: Existing CCD Generation Tools

S. No.	Name of Tool	Advantages
1	Mirth Connect [4]	Open Source with Commercial Offering for support
		Generic Database Interface
		CCD can be implemented using Open Source version but compatibility to other EMR standards is unknown
2	MDHT API	Completely Open Source
		Convenient Interface for CDA
		Model Driven Architecture (UML)
		Can be considered for further development
3	Mule ESB [20]	Open Source with Commercial Offering for modules
		Complete HL7 support only in Commercial Offering
		No inbuilt support for other standards

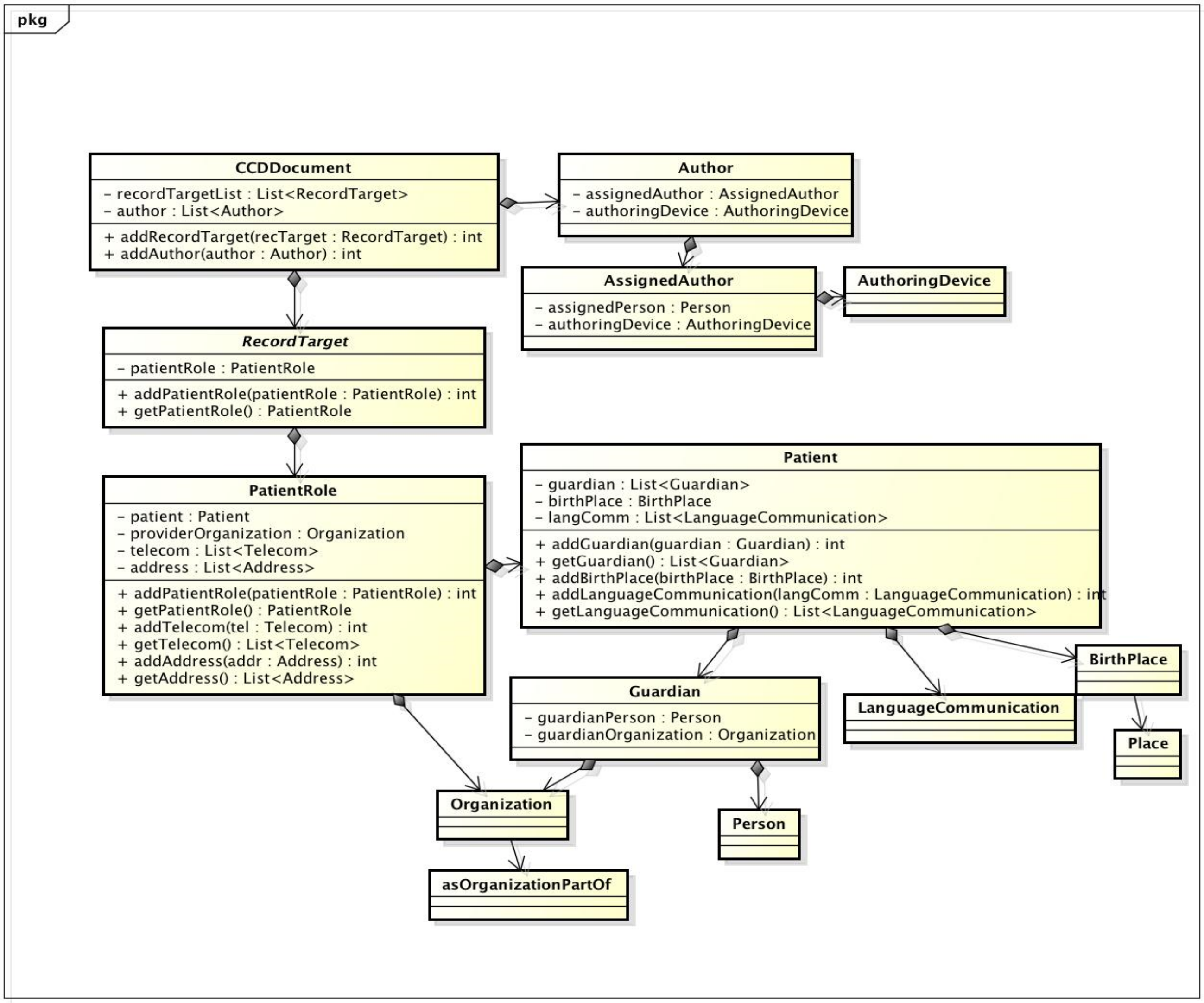

Figure 5: Class diagram for patient profile generation for CCD

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